# SO cross-cl pipeline: decision tree v3 — Bayesian search + profile scheme

*left: search for moments with the evidence · right: profile scheme, every model at its own r*

## A · Bayesian search for moments (evidence, r free)

run BBPower without moments → $M_{0,r}$

is $\chi^2$ good? (PTE > 0.01)
- No → Res_wr: residuals of $M_{0,r}$ all parameters fixed at their best fit (r too); test the d and s families of moments, each with its own β free
- Yes → is r compatible with zero at 1σ?
  - No → Res_wr: residuals of $M_{0,r}$ all parameters fixed at their best fit (r too); test the d and s families of moments, each with its own β free
  - Yes → Yes

moment detected? (*) 0 outside the 68% HPD of an amplitude
- Yes → $M_{1,r}$ = $M_{0,r}$ + a detected family, in order of residual evidence (**); full fit (all parameters free), pruned (***)

is $M_{1,r}$ preferred (****) over $M_{0,r}$? (ln B > 0)
- Yes → $M_{0,r}$ = $M_{1,r}$, start again
- No → Res_r0: residuals of $M_{0,0}$ (r = 0) all parameters fixed, own β free; test the families of moments d, s, dT, d2, s2, ds1, ds2

moment detected? (*) 0 outside the 68% HPD of an amplitude
- Yes → $M_{1,r}$ = $M_{0,r}$ + each detected family, in order of residual evidence (**); full fit (all parameters free), pruned (***)

is one $M_{1,r}$ preferred (****) over $M_{0,r}$? (ln B > 0)
- Yes → $M_{0,r}$ = $M_{1,r}$, start again
- No → $M_1$ = $M_{0,r}$ + combinations of the families tried, from the simplest: the first preferred

is one $M_{1,r}$ (r = 0) preferred over $M_{0,r}$? (ln B > 0) (****)
- No → is r of $M_{0,r}$ compatible with zero at 3σ?
  - No → r detection
  - Yes → upper limit
- Yes → upper limit

Bayesian conclusion: posterior of $M_{0,r}$, or of $M_{1,r}$ if $M_{1,0}$ is preferred

(*) all the amplitudes of the family are fitted (emcee); detected if 0 is outside the 68% HPD (highest posterior density interval) of at least one of them; only the amplitudes with 0 outside their HPD are kept

(**) residual evidence: ln Ẑ of the residual fit of each detected family; it only sets the order in which the families are tried

(***) amplitudes with 0 in their 68% HPD (posterior of the full fit) removed and the model refitted, until all are detected; the pruned model is kept if ln B(pruned vs full) > 0 and is the one compared with $M_{0,r}$

(****) ln B within 2σ of the threshold (A) or between the two best models (B): PolyChord repeated with other seeds, mean of ln Ẑ, up to 3 runs per fit

*then profile scheme*

## B · Profile scheme: every model at its own r

**M* = final model of the tree**
tree chose $M_{0,r}$: candidates c = M* + every combination of the families detected in its last Res_r0 (moments added)
tree chose $M_{1,r}$ ($M_{1,0}$ preferred): c = every model between $M_{1,r}$ and $M_{0,r}$ (moments removed)

the opposite direction of the tree: $M_{0,r}$ and $M_{1,r}$ are the lower and the upper end of the moments

**profiles in r**
M* → r* = its minimum
every candidate c → $r^*_c$ = its minimum

profile: r fixed on a grid, all the other parameters re-fitted (max likelihood)

**compare all the models, each at its own r**
M* at r*, every candidate c at $r^*_c$:
local evidence (****) of each (its moment spectra free; β_d and β_s free in all, T_d only with dT)

**final model = the one with the highest evidence**
(M* itself if no candidate beats it)

**profile of the final model in r**
r_min = its minimum (r* or $r^*_c$);
posterior of r = exp(−Δχ²/2) × prior of the tree

**other parameters**
posterior conditional on r = r_min:
emcee with r fixed, the rest free

0 within the central 3σ interval of the posterior of r?
- Yes → upper limit
- No → r detection

r_95 of the posterior with r ≥ 0

median and 68% of the posterior of r